CMR ENGINEERING COLLEGE

UGC AUTONOMOUS

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III-B.TECH I-SEM (R22) II-MID Examinations-November-2025 **Date: 19.11.2025**

Subject: Artificial Intelligence **Time: 01:30 to 03:30 PM**

Branch: CSE, IT, CSD **Marks: 30 M**

Answer all Questions in Part-A & Answer any FOUR Questions in Part-B

PART-A **5x2=10M**

1. What do you mean by Knowledge Representation? . (CO3) BL2

2. Define First Order Logic. (CO3) BL1

3. Define the Classical Planning Approach. (CO4) BL1

4. Define Ontology Engineering? (CO4) BL1

5. Define inference and Independence. (CO5) BL1

PART-B **4x5=20M**

6. Elucidate Techniques of Knowledge Representation in AI (CO3) BL4

7. What is uncertainty? Explain Inference using Full Joint Distributions (CO4) BL3

8. Define Basic Probability. Explain about Baye's theorem with an example (CO5) BL3

9. Explain in detail the following, (CO4) BL2

   a) Mental Events and Mental objects.

   b) Unification and Lifting

10. Describe Bayesian Networks with an example. (CO5) BL3

11. Differentiate between Propositional and First Order Inference with examples. (CO3) BL4